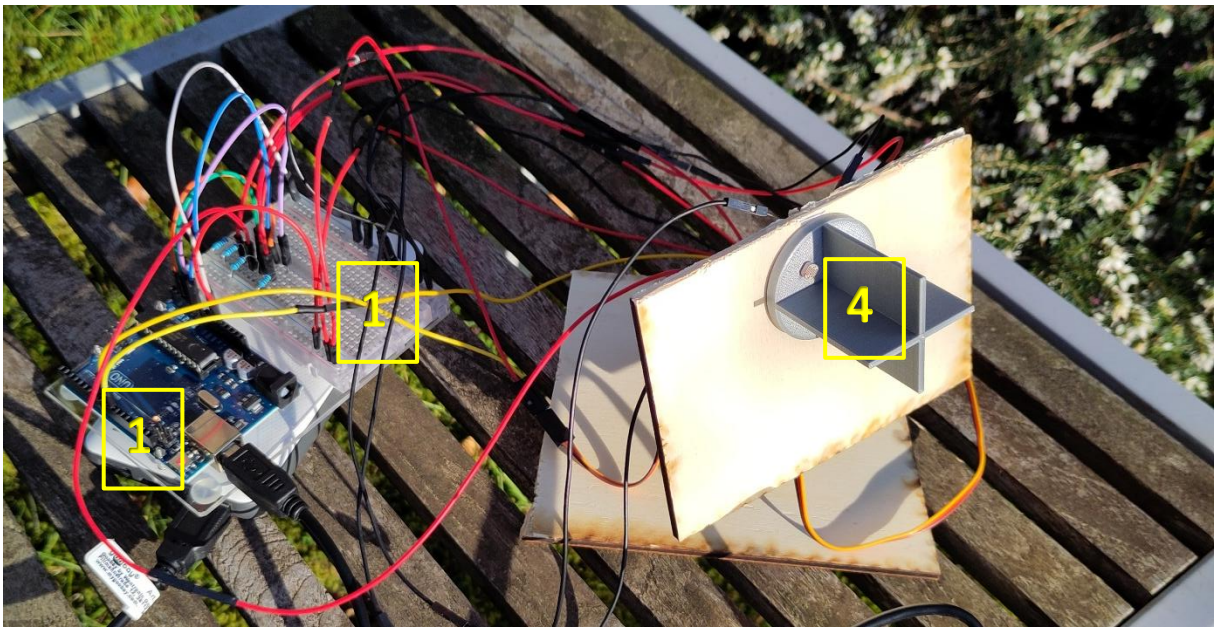
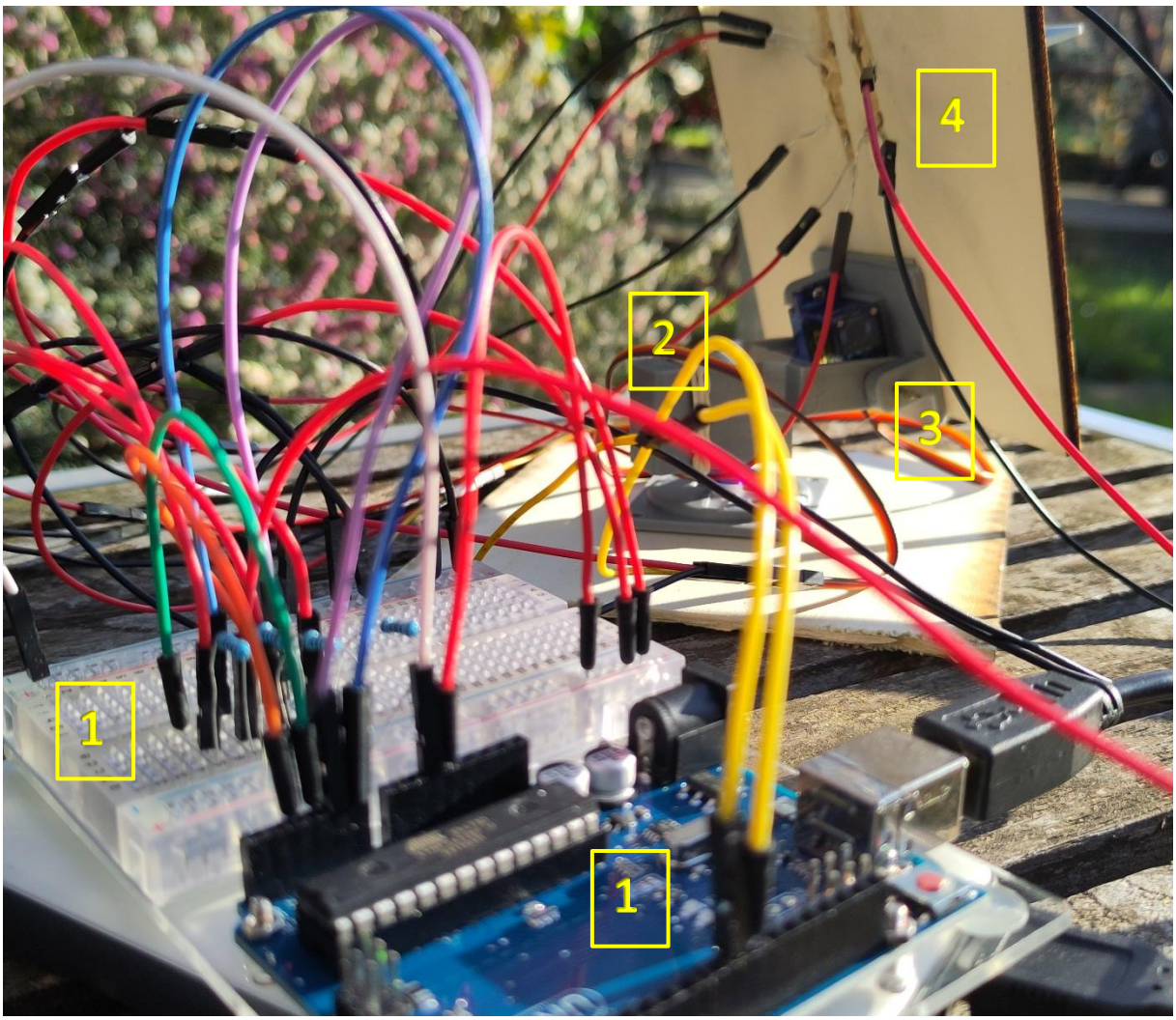
Build a solar tracker

Assembly instructions

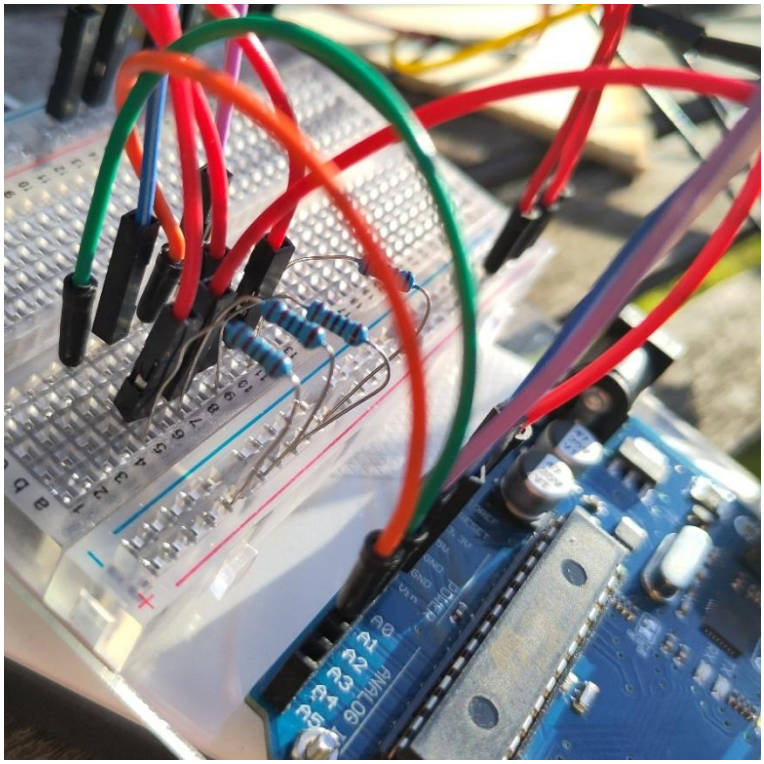




Step 1 – Assemble the separate components

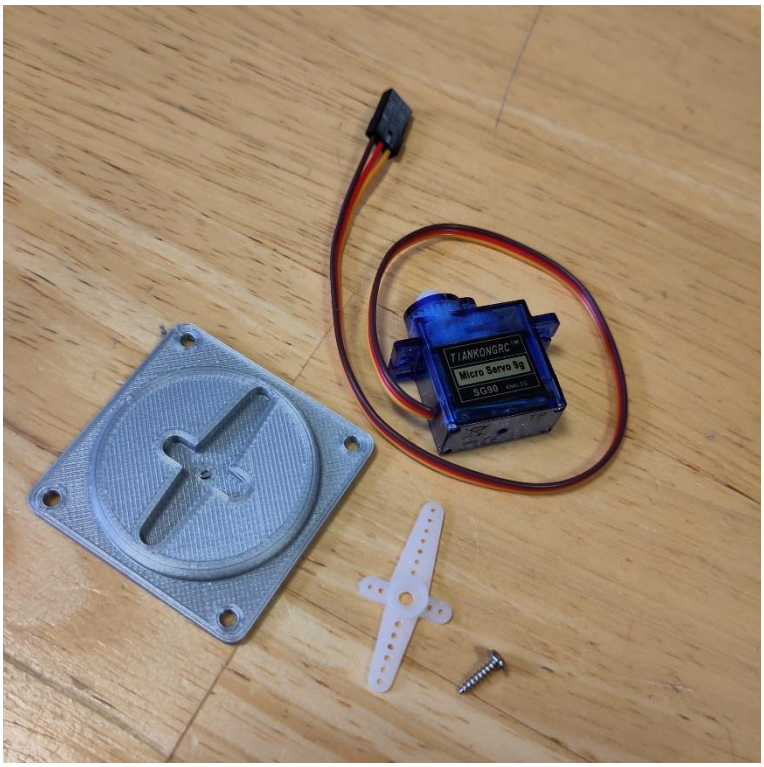
The 4 components can be assembled separately, this work can be divided between the group members. Look closely at the images in this document and the real life example.

Component 1 – Arduino and breadboard



- Remove the protective film from the transparent plate.
- Screw the Arduino board to the transparent plate, point the nut to the top side of the Arduino. Don't forget the plastic spacers between transparent plate and Arduino.
- Stick the rubber feet to the lower part of the transparent plate.
- Attach the breadboard next to the Arduino.

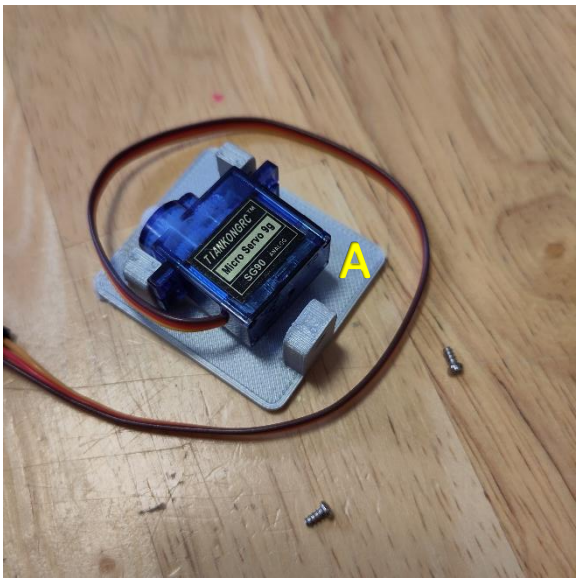
Component 2 – Base with horizontal servo

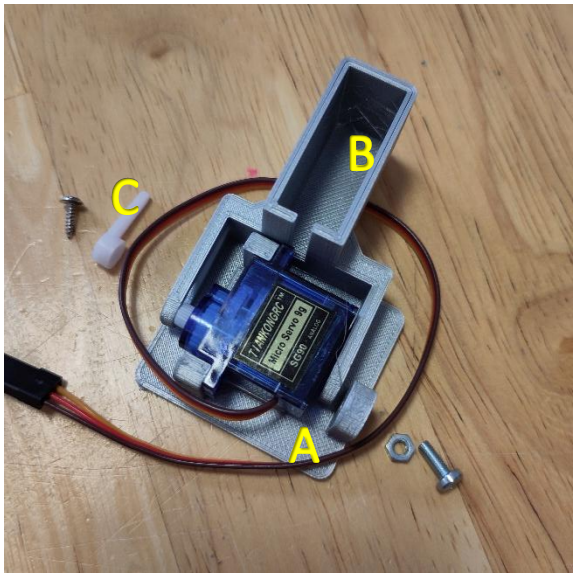


- Assemble the first servo to the base plate using the parts shown in the image.
- This servo will take care of the horizontal movement of the solar tracker.
- Use the long screw that comes with the servo and attach it via the bottom side of the base plate.

Component 3 – Vertical servo

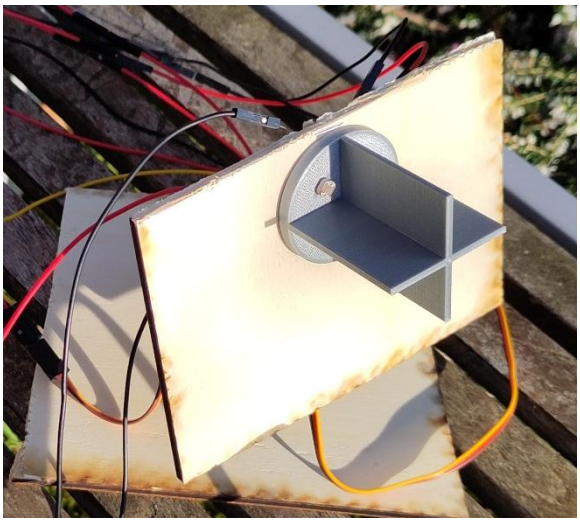
- Connect the servo using the parts shown on the left image. Use 2 short screws to attach the servo to the part A.
- Connect the axis of the servo through the hole in part B to the white one-sided grip C as shown on the right image. Use a long screw.
- Connect the 2 printed parts A and B with the bolt and nut provided (right image).
- Make sure you can still disconnect the servo, we'll have to adjust the servo position later on.

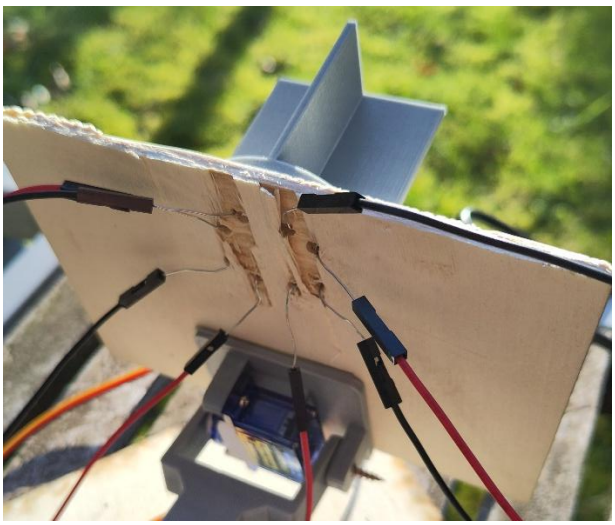




Component 4 – Collimator

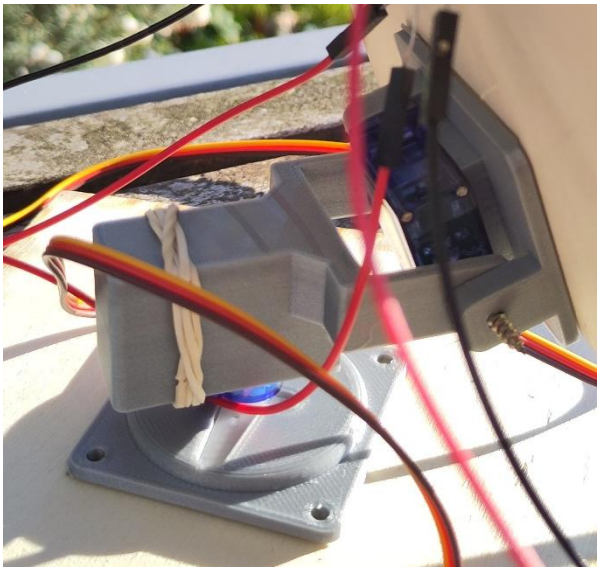
- Put the 4 LDRs through the holes in the collimator.
- Put the LDR connectors through the wooden panel.
- Stick the collimator to the panel using double-sided tape.
- Fold the LDR leads away from each other to avoid contact.
  These will be used later to connect to the breadboard.

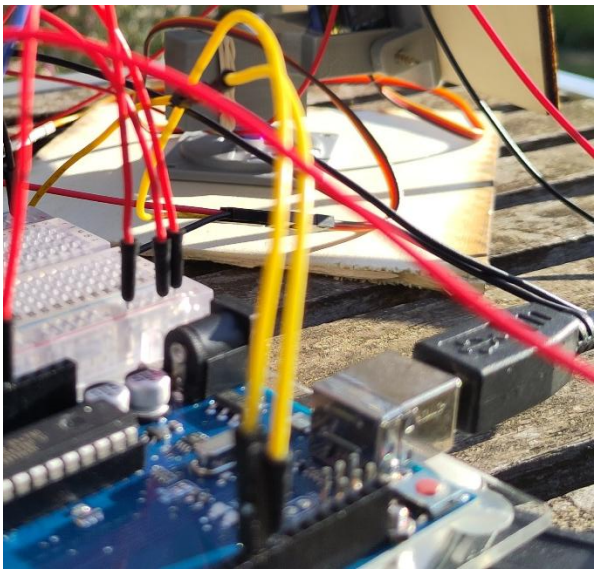




Step 2 – Assemble and connect the servo components

- Connect the vertical servo (component 3) to the base servo (component 2).
- Add a rubber band to fasten the component.
- Connect both servo's to the Arduino using black – red – yellow wires.
  Horizontal servo goes into pin 9, vertical servo into pin 10.
  Use male-male connection wires of sufficient length or use multiple wires.

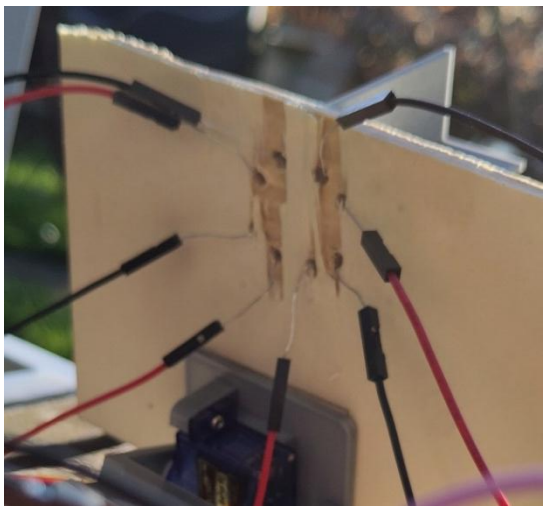


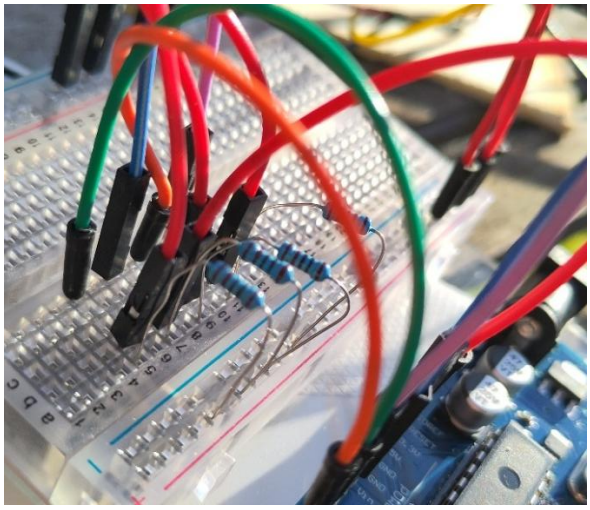


Step 3 – Assemble and connect the light-dependent resistors (LDR)

For each of the 4 LDRs

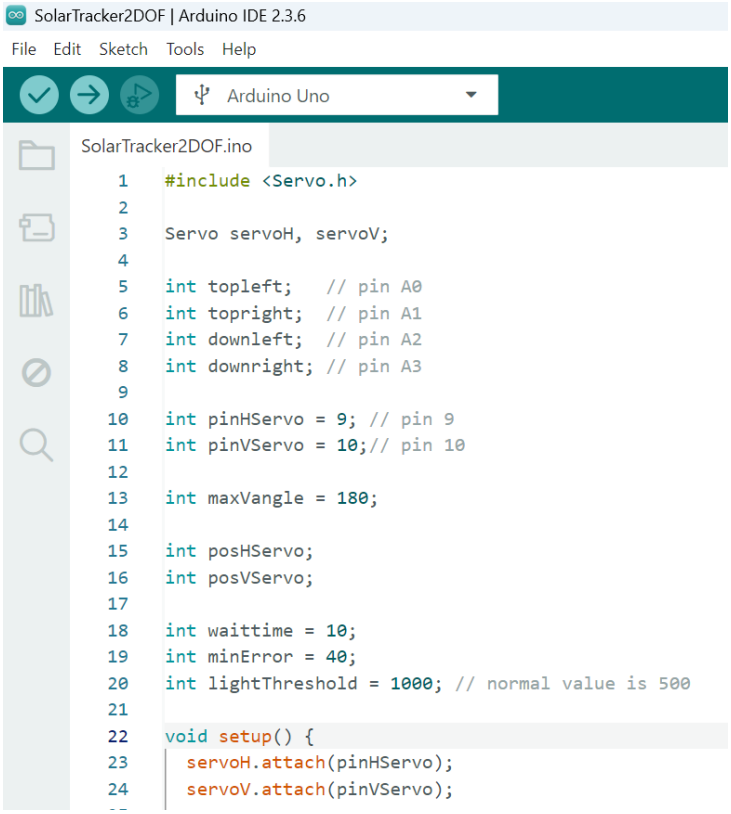
- Attach a black wire to one connector of the LDR as shown on the left image.
- Connect the other end of the black wire to GND on the breadboard.
  Make sure GND of the breadboard is connected to GND of the Arduino.
- Attach a red wire to the other connector of the LDR as shown on the left image.
- Connect the other end of the red wire to a column of the breadboard (see right image).
- Add a 1kΩ resistor between VCC (=5V) and the column of the breadboard where the red wire is connected (see right image). Make sure VCC is connected to VCC of the Arduino.





Step 4 – Upload the steering code and test

- Put the solar tracker out of bright light. The solar tracker will only move when there is sufficient light.
- Download the code *SolarTracker2DOF.ino* from github; **https://github.com/roel31415/SolarTracker**
- Use the Arduino IDE to upload the code to the Arduino board.
- The solar tracker will move to its starting position (which is 90 degrees on both servos). Make sure the wooden plate holding the collimator is positioned vertically. You will have to re-adjust the vertical servo to do so. Be careful when doing this as not to break the printed material.
- Now you can test the setup with bright light from your smartphone or direct sunlight.




```
#include <Servo.h>

Servo servoH, servoV;

int topleft;   // pin A0
int topright;  // pin A1
int downleft;  // pin A2
int downright; // pin A3

int pinHServo = 9; // pin 9
int pinVServo = 10;// pin 10

int maxVangle = 180;

int posHServo;
int posVServo;

int waittime = 10;
int minError = 40;
int lightThreshold = 1000; // normal value is 500

void setup() {
  servoH.attach(pinHServo);
  servoV.attach(pinVServo);
```

Step 5 – Attach the solar panel and make it light a LED

- Connect the + (red) and – (black) of the solar panel to the breadboard using separate columns.
- Add a LED with a 220Ω resistor.